

Yash Rawal

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Professional Summary

Applied AI & Machine Learning Engineer serving a **1M+ user scale** by architecting enterprise agentic platforms, hierarchical RAG architectures, and automated intelligence pipelines across 120+ organizations. Specialized in Python, LLM orchestration (LangChain, LangGraph, ReAct loops), vector search, and cloud AI infrastructure (Azure, AWS, GCP). Active contributor to Spring AI, bridging generative AI and model fine-tuning with resilient production systems.

Skills

Languages & Backend: Python, SQL, Java, FastAPI, Spring Boot, PostgreSQL, REST APIs

AI & ML Engineering: Agentic AI, Tool RAG, Multi-Agent Systems, RAG & Vector Search, Context Graphs

AI & ML Frameworks: LangChain, LangGraph, Spring AI, PyTorch, vLLM, LlamaIndex, GROBID

Platforms & Models: Azure, AWS, GCP, Azure OpenAI, Amazon Bedrock, Meta Llama, Claude, Qwen

Infrastructure & Security: Docker, Kubernetes, Linux, CI/CD, LLM-as-Judge, Red Teaming, PII Sanitization

Work Experience

Bigscal Technologies Pvt. Ltd.
AI/ML Engineer Oct 2024 - Present
Surat, India

- Deployed a production-scale agentic platform and multi-agent systems serving **1M+ active ERP users** across 120+ enterprise tenants on cloud infrastructure (Azure, AWS, GCP).
- Architected and deployed multi-agent workflows with LangChain and LangGraph, utilizing **MCP Server Tool Routing (Tool RAG)** and A2A communication with integrated guardrails.
- Designed a capability contract layer to execute nested profiles inside a single ReAct loop, reducing execution overhead by **3x** and token consumption by 70%.
- Implemented a hierarchical RAG pipeline indexing 10,000+ technical documents using GROBID, pdfplumber, and PostgreSQL/PGVector, achieving a **90%+ Recall@10** retrieval rate.
- Engineered an automated LLM evaluation and red-teaming control plane to log reasoning trajectories, scoring agent behaviors via LLM-as-a-judge and **RLHF preference datasets**.
- Fortified agentic workflows against OWASP LLM vulnerabilities with phantom-token context isolation, runtime tool telemetry, and **automated PII redaction**.

Bigscal Technologies Pvt. Ltd.
AI/ML Engineer Intern Apr 2024 - Sep 2024
Surat, India

- Engineered clinical summarization pipelines for 116K+ medical records with 4-bit quantized Llama 3, and designed real-time WebSocket audio streaming for Indic ASR with **NeMo RNN-T and CTC decoding**.
- Researched audio digital signal processing (STFT, Mel-spectrograms, MFCC feature extraction via librosa/torchaudio) and CNN audio architectures (VGGish, ResNet), compiling custom Indic LJSpeech-format datasets.

Projects

Modular Agent Harness & Memory System ("Brain") | Open Source
Principal Creator & Architect of an open-source plug-and-play agentic framework designed as modular building blocks for flexible enterprise workflows, implementing a formal ReAct loop with step budgets, vector tool routing, and dynamic MCP settings caching.
Java, Spring Boot, PostgreSQL, WebSockets

Spring AI (spring-projects/spring-ai) | Open Source
Merged PR #5711 to propagate reasoningContent from AWS Bedrock and OpenAI-compatible models, and resolved Converse API prompt-caching TTL latency consistency.
Java, Spring Boot, Python

Mini Language Model (~85M Parameter Architecture) | Personal Research
Pretrained an ~85M-parameter transformer to ~1.43 validation loss on TinyStories targeting **edge hardware deployment**; engineered FragmentStream chunked attention to eliminate $O(T^2)$ VRAM overhead and custom Triton kernels with KV caching for low-latency inference.
Python, PyTorch, CUDA, Triton, Tesla P100/T4

Education

Master of Computer Applications (Artificial Intelligence & Machine Learning) - Parul University 2022 - 2024

Certifications

Microsoft Certified: Azure AI Fundamentals (AI-900)	Microsoft
Microsoft Certified: Azure Fundamentals (AZ-900)	Microsoft